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HONEYCHAIN: EVIDENCE-BACKED HONEY TRUST WITH SEALED-SAMPLE AUDITS AND SMART HIVE MONITORING



- **Problem Statement ID – SIH26021**
- **Problem Statement Title - HONEY CHAIN: A Blockchain-based system for honey traceability and smart beekeeping management**
- **Theme - Agriculture, FoodTech & Rural Development**
- **PS Category - Software**
- **TEAM ID - 189024**
- **Team Name - TEAM ICONICS**





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HONEY CHAIN

Evidence-Backed Honey Provenance & Authenticity Auditing

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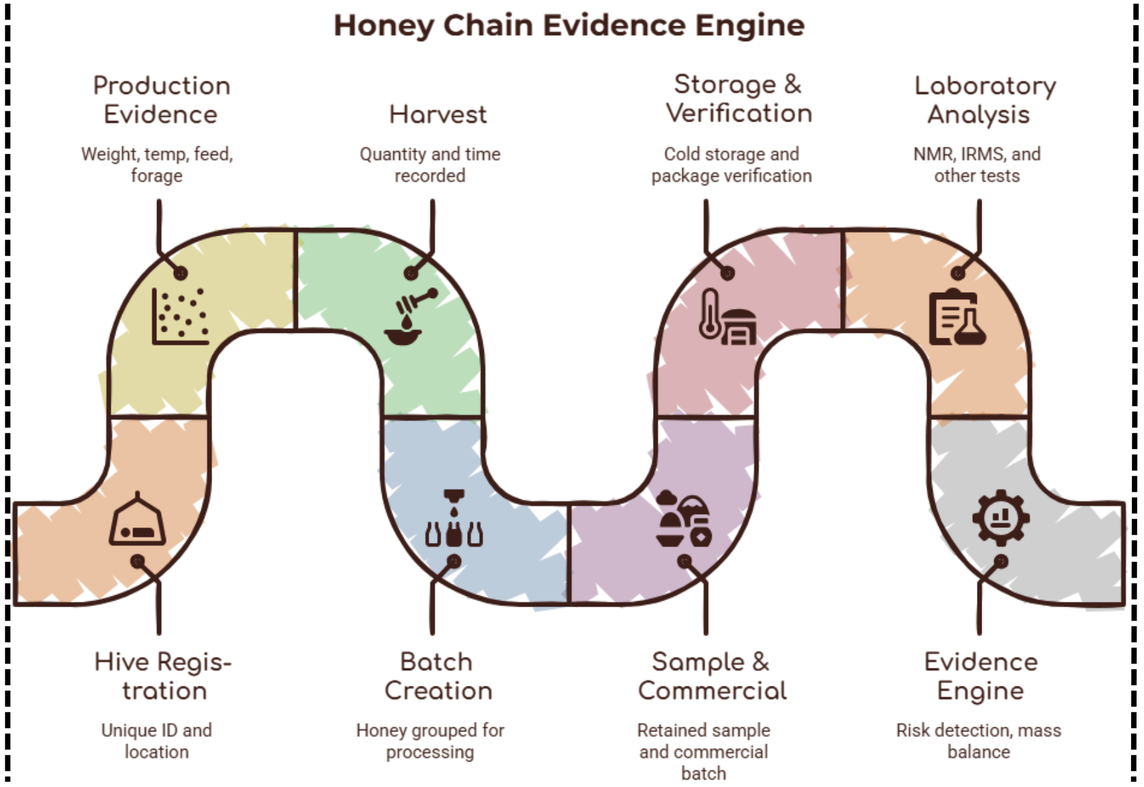
REAL-WORLD PROBLEM

- Unverifiable honey origin
- Post-harvest adulteration & substitution
- India's missing reference data
- Limited laboratory capacity
- Syrup feeding

HOW IT ADDRESSES THE PROBLEM

- Counterfeit honey → QR-based authenticity & batch verification.
- No traceability → Tracks honey from Hive → Batch → Lab → Jar → Consumer.
- Poor quality assurance → Links moisture/FSSAI lab reports to each batch.
- Low beekeeper income → Direct access to a verified marketplace.

IDEA / SOLUTION



WOW FACTOR

- REFERENCE STATIONS at Universities / ICAR-KVK sites / documented crop fields
- Anti-substitution protection
- Indian honey reference dataset
- Bee-keeping management system

RESULT :

"Each sample is bound to its feed and hive record, tested by an accredited lab, and linked to the jar through a tamper-evident NFC tag. The result is evidence you can audit, and a reference dataset India doesn't have."

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TECHNICAL APPROACH

HONEY CHAIN'S EVIDENCE INFRASTRUCTURE FOR HONEY AUDITING

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Hive scale + feeder scale + temp/humidity

Data capture from hives and feeders

LoRa (field) / Wi-Fi (station)

Data transmission to gateway

Gateway (Raspberry Pi)

Data processing and storage

Backend API (FastAPI)

Data analysis and mass-balance calculation

Merkle batcher

Smart contract creation

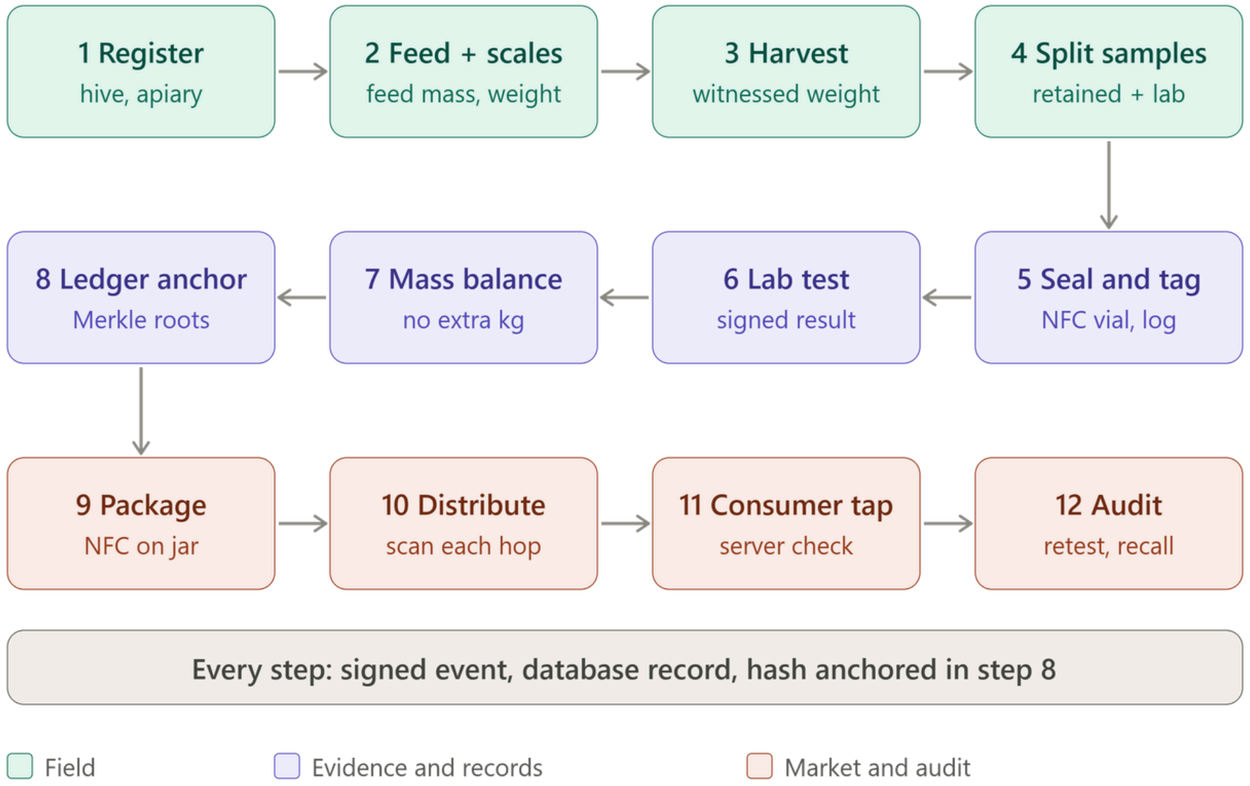
Apps: Beekeeper app, Lab/regulator dashboard, Consumer verification page

User interface for data access

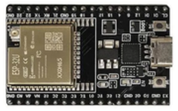
NFC tap (NTAG 424 DNA TagTamper)

Consumer verification

WORKFLOW



TECH STACK:

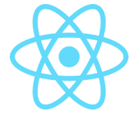






FastAPI PostgreSQL HYPERLEDGER BESU





Flutter

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HONEY CHAIN: PROTOTYPE SCREENS

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My hives

- Hive #12 +1.8 kg
- Hive #9 +1.1 kg
- Hive #7 Mite risk
- Hive #3 +0.9 kg

AI alert: mite risk in Hive #7. Inspect.

Hives Batches Market

1. Hive monitor

Create batch

Batch ID **HC-0421**

Apiary **Kanchipuram**

Lab report **Attached**

Hash 9f3a...c21e stored on ledger

Print QR label

2. Batch and QR

Scan result

OK

Authentic honey

Batch **HC-0421**

Origin **Kanchipuram, TN**

Harvested **14 Sep 2026**

Moisture **17.8% (pass)**

- Hive - verified
- Lab report - verified
- Jar sealed - verified
- Store shelf - verified

3. Consumer verified

Scan result

X

Tampered or fake

Batch **HC-0999**

Status **Hash mismatch**

Failed at **Step 3 (lab)**

This jar cannot be verified.
Do not buy. Report it.

Report to FSSAI

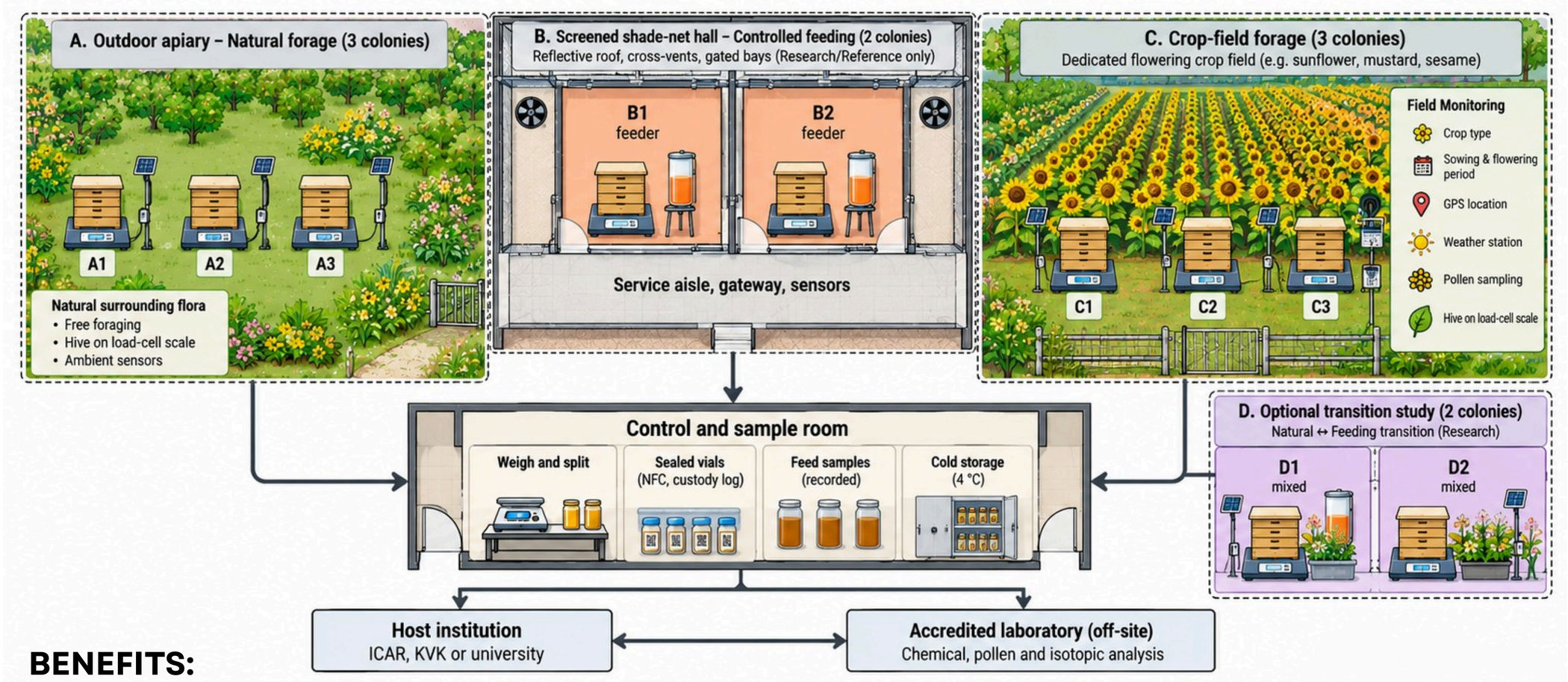
4. Tamper detection

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HONEY CHAIN

BUSINESS MODEL : REFERENCE HONEY STATION (WOW FACTOR)

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Total Colonies **10** (Prototype)

Main Production Natural Forage & Crop Field (A + C)

Feeding Only for Research (B) (Not for production)

Monitored, Sampled & Lab Verified

BENEFITS:

- **A national reference-data asset :** Builds India's open, sensor-linked reference honey dataset**.**
- **Institutions adopt it, so it scales without a new platform:** Hosted by ICAR, KVKs and universities, and plugged into Madhukranti and NBHM. Accredited labs get provenance-known samples, and each new host site adds data at low cost.
- **Farmers and honey businesses gain trusted proof**: Crop-field trials show which crops sustain colonies, guiding farmer,beekeeper forage planning. Audit-ready origin evidence supports premium pricing and export confidence.

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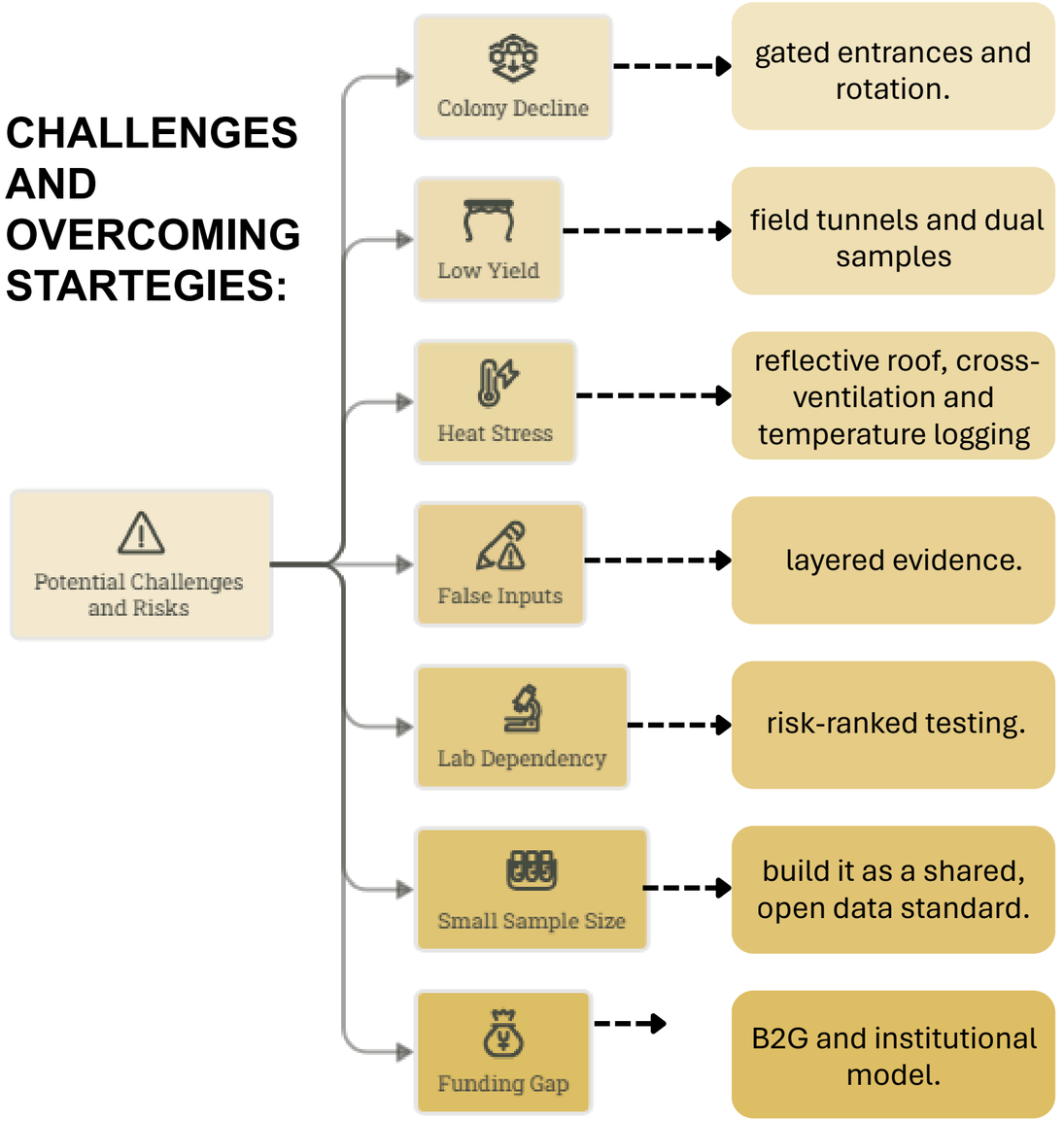
FEASIBILITY AND VIABILITY

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FEASABILITY ANALYSIS:



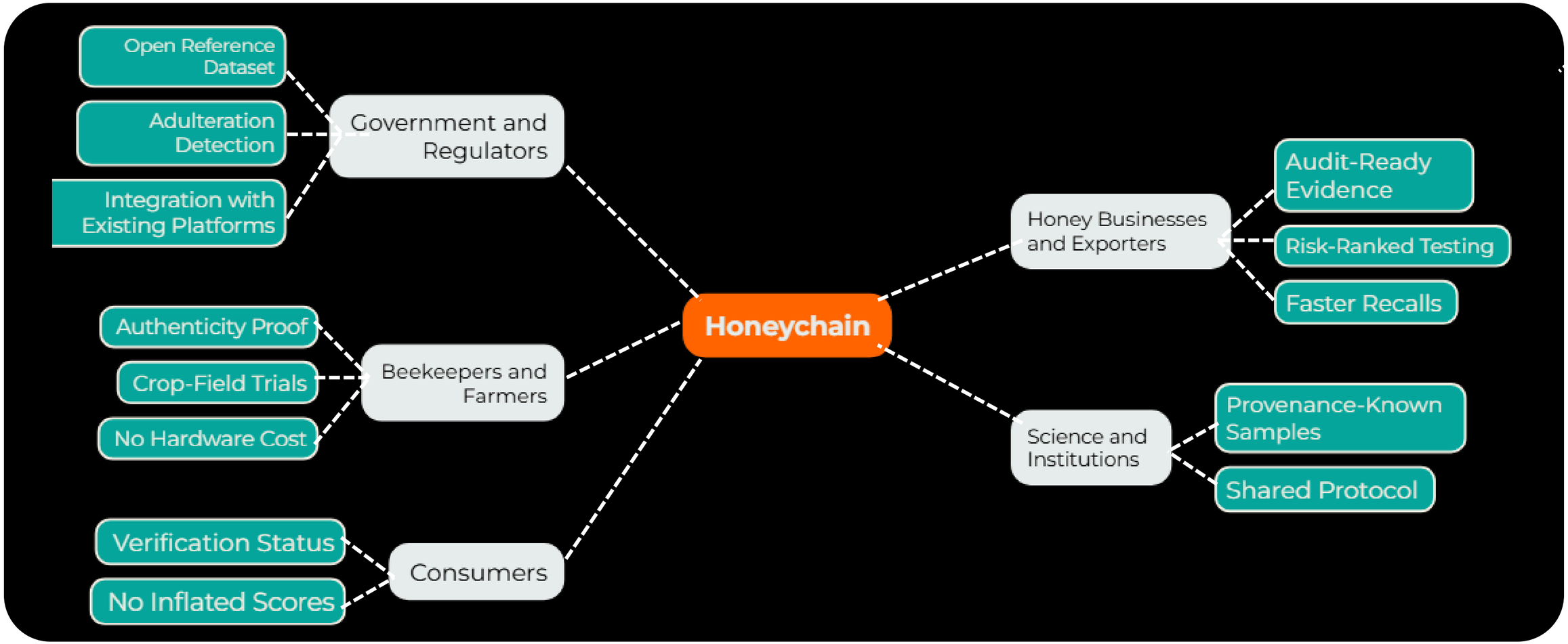
CHALLENGES AND OVERCOMING STARTEGIES:



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IMPACT AND BENEFITS

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- Eradicating 70% adultration crisis
- Higher Export value
- Safeguarding 10,000+ crore in crop pollination value
- FPO - fair trad margins

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